

Physics-Informed AI Framework for Structural Failure Prediction in Rotating Disk Systems

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Abstract

Structural failure in high-speed rotating disk systems is a major concern in aerospace, automotive, energy storage, and industrial machinery because of the extreme centrifugal stresses generated during operation. Traditional failure assessment methods primarily rely on analytical stress calculations and finite element simulations, which may become computationally expensive for large-scale predictive monitoring applications. This study presents a physics-informed artificial intelligence framework for structural failure prediction in rotating disk systems by integrating classical elasticity-based stress analysis with machine learning techniques.

A synthetic dataset containing 50,000 rotating disk configurations was generated using validated rotating disk stress equations under varying material properties, geometric parameters, and operating conditions. Failure labels were assigned using the Von Mises yield criterion, enabling the creation of a physically consistent supervised learning dataset. The generated dataset included variations in material density, Young's modulus, Poisson's ratio, yield strength, disk geometry, and rotational speed.

Four machine learning algorithms were evaluated for structural failure prediction: Random Forest, XGBoost, Support Vector Machine (SVM), and Multi-Layer Perceptron Neural Network. The dataset was divided into training and testing subsets using an 80–20 split, and feature normalization was applied where required. Among all investigated models, the SVM classifier achieved the highest performance with an accuracy of 99.49%, precision of 99.49%, recall of 99.47%, and F1-score of 99.48%. The Neural Network, XGBoost, and Random Forest models achieved accuracies of 99.23%, 98.87%, and 97.87%, respectively.

Feature importance and model interpretability analyses revealed that rotational speed, material strength, and geometric parameters are the dominant factors influencing structural failure prediction. The strong agreement between physics-based failure criteria and machine learning predictions demonstrates the effectiveness of integrating analytical mechanical modeling with artificial intelligence.

The proposed framework provides a computationally efficient and physically interpretable approach for intelligent structural health monitoring and predictive maintenance of high-speed rotating mechanical systems. The results highlight the significant potential of physics-informed machine learning for next-generation engineering reliability and failure prediction applications.

Keywords—Physics-Informed Machine Learning, Rotating Disk Mechanics, Structural Failure Prediction, Predictive Maintenance, Structural Health Monitoring, Support Vector Machine, Engineering Reliability, Intelligent Monitoring Systems

1. Introduction

Rotating disk systems are essential components in many engineering applications, including gas turbines, flywheels, brake rotors, aerospace propulsion systems, power generation equipment, and high-speed industrial machinery [1], [2]. During operation, these systems experience large centrifugal forces generated by high angular velocities, leading to complex radial and circumferential stress distributions within the disk. Under extreme operating conditions, excessive stress concentration, thermal loading, material fatigue, and crack initiation can result in catastrophic structural failure, creating serious safety concerns and significant economic losses in critical mechanical systems [3], [4].

Conventional approaches to rotating disk failure analysis primarily rely on analytical stress calculations, experimental testing, and finite element simulations [1], [5], [12]. Classical elasticity theory provides important insight into radial and hoop stress behavior in rotating components, but these methods often become computationally expensive when applied to large-scale parametric studies or real-time structural monitoring. In addition, traditional techniques may struggle to provide adaptive failure prediction under dynamically changing operating conditions where multiple mechanical parameters simultaneously affect structural integrity.

In recent years, artificial intelligence and machine learning techniques have gained increasing attention in predictive maintenance and structural health monitoring applications [6], [7]. Data-driven learning algorithms can identify hidden patterns in operational data and detect abnormal structural behavior before catastrophic failure occurs. Machine learning models such as Random Forests, Support Vector Machines, Gradient Boosting algorithms, and Deep Neural Networks have shown strong potential in fault diagnosis, vibration analysis, and predictive maintenance across modern industrial systems [8], [9], [10], [11].

Despite these advances, many existing AI-based prediction models function as black-box systems and often fail to incorporate the underlying physical principles governing mechanical stress evolution. The absence of physics-based constraints can reduce both interpretability and reliability, particularly in safety-critical engineering applications. As a result, integrating classical mechanical modeling with artificial intelligence has become an important research

direction for improving predictive accuracy while preserving engineering interpretability [23], [24].

In this work, a physics-informed artificial intelligence framework is proposed for predicting structural failure in rotating disk systems. The methodology combines classical rotating disk stress analysis with machine learning-based predictive modeling to identify unsafe operating conditions and estimate structural instability under varying mechanical parameters. Radial and hoop stress distributions are generated using analytical formulations across different angular velocities, material properties, and simulated defect conditions. The resulting dataset is then used to train machine learning models capable of predicting structural failure tendencies in high-speed rotating disks.

2. Literature Review

2.1 Structural Behavior of Rotating Disk Systems

Rotating disk systems are widely used in aerospace turbines, flywheel energy storage systems, automotive brake rotors, compressors, and high-speed industrial machinery because of their ability to operate at extremely high rotational speeds [1], [2], [3]. During operation, these disks are subjected to large centrifugal forces that generate complex radial and circumferential stress distributions throughout the structure. As angular velocity increases, stress concentrations within the disk can exceed material limits, leading to deformation, crack initiation, fatigue damage, and ultimately catastrophic structural failure [3], [4], [14].

Early studies of rotating disks were primarily based on classical elasticity theory, which provided analytical solutions for radial and hoop stress distributions under different geometric and material conditions [1], [5]. These analytical models established the theoretical foundation for understanding stress evolution in rotating structures and continue to play an important role in mechanical design and safety evaluation. However, many traditional analytical approaches assume simplified operating conditions and may not fully capture the nonlinear failure mechanisms encountered in practical engineering applications.

The development of computational techniques such as Finite Element Analysis (FEA) significantly improved structural analysis capabilities by enabling detailed visualization of stress distributions and deformation behavior under realistic operating conditions [12], [13]. Numerous studies have used FEA to investigate crack propagation, thermal stress effects, and fatigue life estimation in rotating disks exposed to high-speed loading environments. Although numerical simulations provide highly accurate structural predictions, repeated simulations across multiple operating conditions can become computationally expensive, limiting their suitability for real-time predictive monitoring systems.

2.2 Structural Failure and Fatigue Mechanisms

Structural failure in rotating disks is commonly associated with stress concentration, cyclic fatigue, thermal loading, material defects, and crack propagation [4], [14], [15]. Among these mechanisms, fatigue-induced crack growth is one of the most critical causes of failure in high-speed rotating machinery because repeated cyclic loading gradually weakens the structural integrity of the disk over time [14], [17].

Fracture mechanics and fatigue analysis have been extensively studied to estimate the operational lifespan of rotating components [4], [14], [15]. Experimental investigations have shown that microscopic cracks formed near stress concentration regions can progressively propagate under repeated rotational loading, eventually resulting in sudden structural fracture [15], [17]. Thermal effects generated by friction and high-speed operation can further accelerate material degradation and structural instability [3], [16].

Several fatigue prediction approaches based on stress-life and strain-life relationships have been proposed to estimate failure initiation and structural durability [14], [17]. While these methods provide valuable engineering insight, they often require extensive experimental calibration and may not adapt efficiently to dynamically changing operating conditions. As a result, increasing attention has been directed toward intelligent predictive methods capable of learning structural degradation behavior directly from operational data [7], [18].

2.3 Artificial Intelligence in Structural Health Monitoring

Artificial intelligence and machine learning techniques have recently emerged as powerful tools for structural health monitoring and predictive maintenance applications [6], [7], [18]. Machine learning algorithms can identify complex patterns within operational datasets and predict abnormal system behavior with high computational efficiency. Algorithms such as Support Vector Machines (SVM), Random Forests (RF), Artificial Neural Networks (ANN), Gradient Boosting methods, and deep learning architectures have been widely applied in fault diagnosis and predictive maintenance research [8], [9], [10], [11].

In mechanical engineering systems, AI-based methods have been used for vibration analysis, bearing fault detection, gearbox diagnostics, crack identification, and fatigue prediction [7], [18], [19]. These approaches typically rely on sensor-derived information such as vibration signals, acoustic emissions, temperature variations, and stress measurements to classify structural conditions and estimate remaining useful life.

Among existing machine learning techniques, ensemble-based methods such as Random Forest and XGBoost have demonstrated particularly strong predictive performance although the Support Vector Machine achieved the highest classification accuracy in the present study because of their robustness, computational efficiency, and ability to model nonlinear engineering datasets [8], [10], [21]. Deep learning approaches, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, have also shown promising results in time-series fault prediction tasks [11], [20]. However, many AI-based

systems operate as black-box models and often lack physical interpretability, limiting their reliability in safety-critical engineering applications.

2.4 Physics-Informed Artificial Intelligence Approaches

Recent advances in intelligent engineering systems have led to the development of physics-informed machine learning frameworks that combine physical laws with data-driven predictive techniques [23], [24]. Unlike purely data-driven AI models, physics-informed approaches incorporate governing equations, stress relationships, and engineering constraints directly into the learning process, improving both reliability and interpretability.

Several studies have demonstrated that integrating physical principles with machine learning can improve prediction stability and reduce overfitting, particularly when available training data are limited [23], [24]. In structural engineering applications, physics-informed AI has been explored for crack growth prediction, material deformation analysis, fluid–structure interaction problems, and fatigue estimation [23], [24].

The incorporation of physical constraints enables machine learning models to maintain consistency with established engineering principles while preserving the flexibility of data-driven learning. Consequently, physics-informed approaches have emerged as a promising research direction for safety-critical engineering applications where both predictive accuracy and physical reliability are essential [24].

Despite the growing interest in physics-informed methodologies, relatively limited research has focused specifically on structural failure prediction in rotating disk systems using hybrid mechanical–AI frameworks. Most existing studies either depend entirely on analytical simulations or employ purely data-driven models without incorporating the underlying mechanical stress behavior into the prediction process.

2.5 Research Gap and Motivation

The reviewed literature indicates that traditional rotating disk analysis methods provide accurate mechanical insight but often become computationally expensive for large-scale predictive monitoring applications [12], [13]. In contrast, purely data-driven machine learning approaches offer strong predictive capability but frequently neglect the physical mechanisms governing stress evolution and structural degradation [20], [21].

Furthermore, only limited work has explored the integration of analytical rotating disk stress formulations with machine learning-based failure prediction frameworks for intelligent structural health monitoring. The lack of physically interpretable AI models therefore represents an important gap in the development of reliable predictive systems for high-speed rotating mechanical components [23], [24].

To address this gap, the present study proposes a physics-informed artificial intelligence framework for structural failure prediction in rotating disk systems by combining classical mechanical stress analysis with machine learning techniques. The proposed approach aims to improve predictive performance while maintaining engineering interpretability and computational efficiency for advanced structural health monitoring applications.

3. Mathematical Modeling of Rotating Disk Systems

3.1 Introduction to Rotating Disk Mechanics

Rotating disk systems are essential components in many high-speed mechanical and aerospace applications, where rotational motion generates substantial centrifugal loading within the structure [1], [2]. During operation, each elemental particle of the disk experiences an outward radial force that increases with angular velocity. As rotational speed rises, internal stresses develop throughout the disk body, leading to deformation, stress concentration, fatigue accumulation, and possible structural instability [2], [3].

The structural behavior of rotating disks is primarily governed by radial stress and circumferential (hoop) stress distributions generated by centrifugal effects. Accurate evaluation of these stresses is critical for predicting failure conditions and designing reliable high-speed rotating systems. In this study, analytical elasticity-based formulations are used to model the stress behavior of a solid rotating disk under varying operating conditions [1], [5].

These assumptions are consistent with classical elasticity formulations commonly used in rotating disk stress analysis [1], [5].

A typical rotating disk geometry and the associated stress distribution are illustrated in Figure 1 [1], [5].

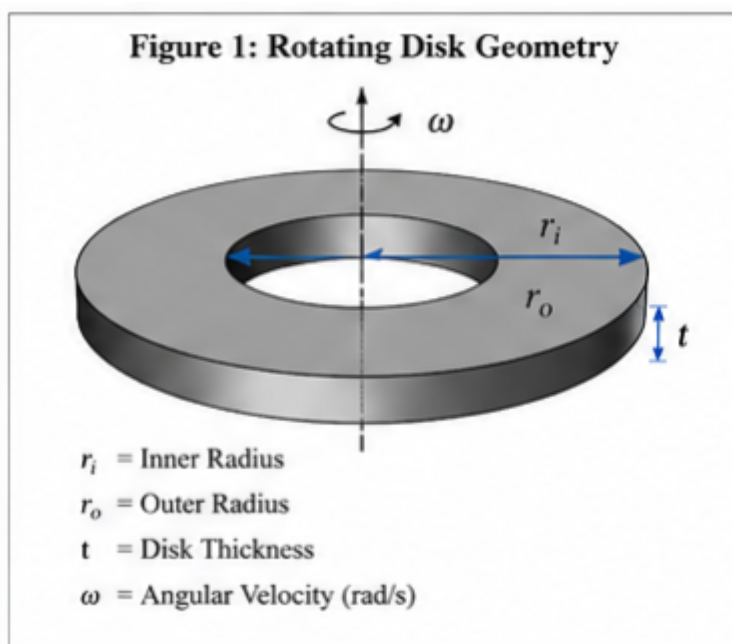


Figure 1. Geometry of a rotating disk showing inner radius, outer radius, thickness, and angular velocity.

3.2 Assumptions Used in the Mechanical Model

To simplify the analytical formulation of the rotating disk system, the following assumptions are adopted:

1. The rotating disk is homogeneous and isotropic.
2. Material properties remain uniform throughout the disk.
3. The disk has constant thickness.
4. Elastic deformation follows linear elasticity theory.
5. Plane stress conditions are assumed.
6. Rotational motion is steady and axisymmetric.
7. Thermal expansion and aerodynamic effects are initially neglected.

These assumptions make it possible to model the stress distribution analytically while maintaining sufficient engineering accuracy for predictive structural analysis.

3.3 Centrifugal Force in Rotating Disks

When a disk rotates with angular velocity ω , each infinitesimal element experiences centrifugal acceleration directed outward from the center of rotation. The centrifugal body force acting on a rotating element can be expressed as:

$$F_c = mr\omega^2$$

The centrifugal force formulation is derived from classical rotational dynamics and elasticity theory [1], [2].

where:

- F_c represents the centrifugal force,
- m is the mass of the rotating element,
- r is the radial distance from the center,
- ω is the angular velocity.

The centrifugal loading generated at high rotational speeds produces internal stress fields that are responsible for structural deformation and potential mechanical failure.

Crack initiation and propagation mechanisms commonly observed in rotating disk systems are illustrated in Figure 2 [4], [14], [15].

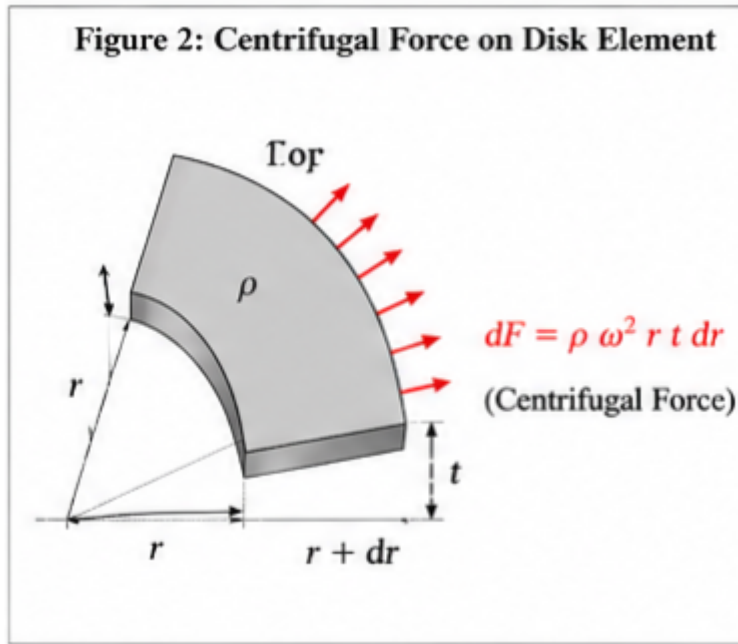


Figure 2. Centrifugal force acting on an elemental section of a rotating disk.

3.4 Radial Stress Distribution

Radial stress develops along the radial direction of the rotating disk as a result of centrifugal loading. Based on classical elasticity theory for solid rotating disks, the radial stress distribution is given by:

$$\sigma_r = \frac{3+\nu}{8} \rho \omega^2 (R^2 - r^2)$$

The analytical expression for radial stress is obtained from classical elasticity solutions for solid rotating disks [1], [5].

where:

- σ_r = radial stress,
- ν = Poisson's ratio,
- ρ = material density,
- ω = angular velocity,
- R = outer radius of the disk,
- r = radial position from the disk center.

The radial stress gradually decreases from the center toward the outer boundary of the disk.

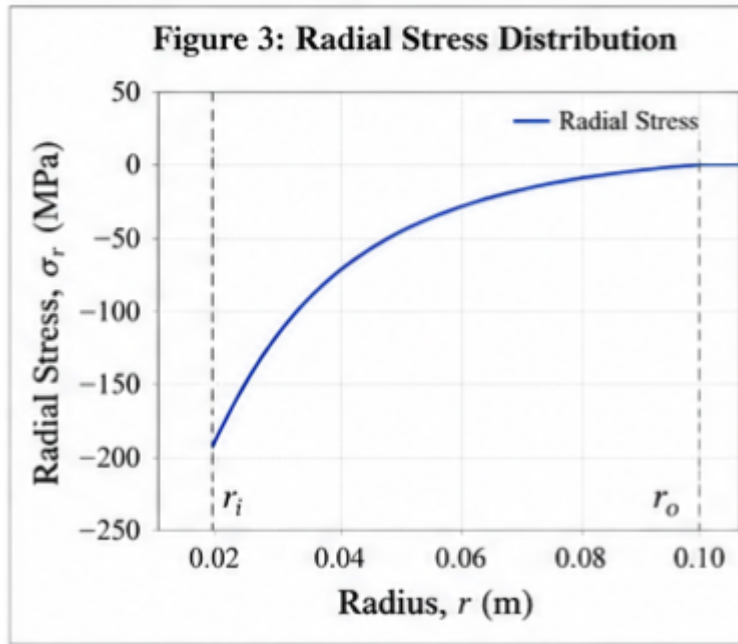


Figure 3. Radial stress variation along the disk radius.

3.5 Hoop Stress Distribution

Hoop stress, also known as circumferential stress, acts tangentially along the circumference of the rotating disk and is generally the dominant stress component responsible for crack initiation and structural failure in high-speed rotating systems.

The hoop stress distribution for a solid rotating disk is expressed as:

$$\sigma_{\theta} = \frac{3+\nu}{8} \rho \omega^2 \left(R^2 - \frac{1+3\nu}{3+\nu} r^2 \right)$$

The hoop stress formulation is based on established rotating disk stress theory and has been widely used in mechanical design and structural safety assessment [1], [5].

where:

- σ_{θ} = hoop stress,
- ν = Poisson's ratio,
- ρ = material density,
- ω = angular velocity,
- R = outer radius,
- r = radial distance from the center.

Hoop stress typically reaches its maximum value near the central region of the disk and plays a major role in fatigue crack initiation under repeated rotational loading.

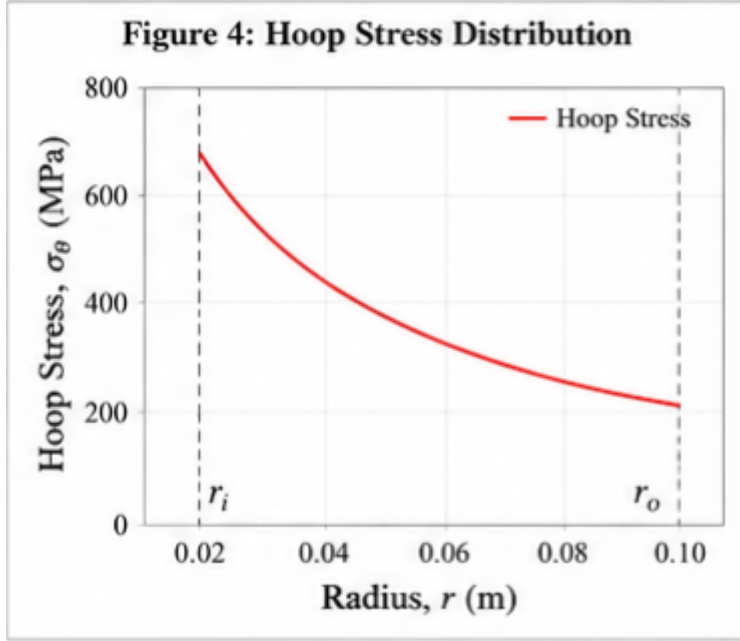


Figure 4. Hoop stress variation across the rotating disk radius.

3.6 Influence of Angular Velocity on Stress Evolution

The analytical stress equations show that both radial and hoop stresses are proportional to the square of the angular velocity:

$$\sigma \propto \omega^2$$

This quadratic relationship between stress magnitude and angular velocity is a fundamental characteristic of rotating disk mechanics [1], [2], [5].

As a result, even small increases in rotational speed can generate significantly larger stress magnitudes within the rotating disk structure. This quadratic relationship highlights the importance of accurate stress monitoring in high-speed rotating systems such as gas turbines, jet engines, flywheel energy storage systems, and industrial rotors.

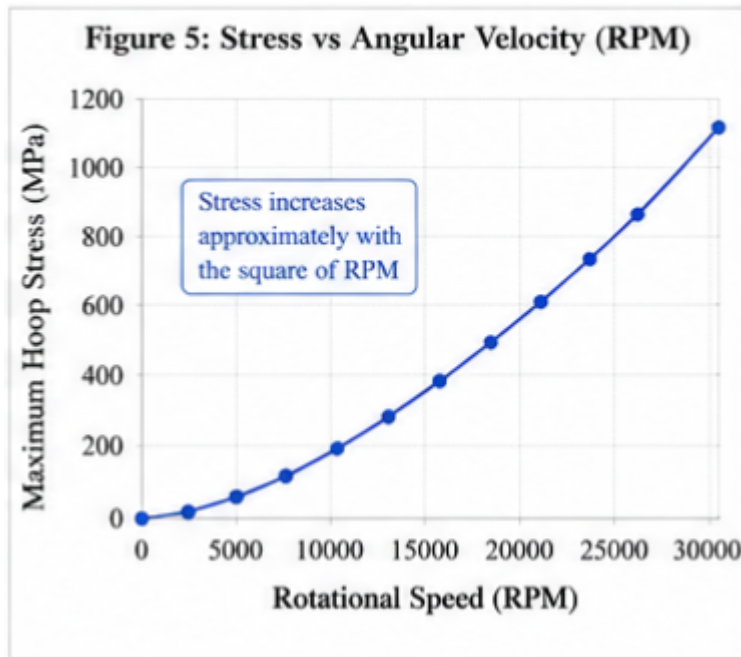


Figure 5. Increase in stress magnitude with rotational speed.

3.7 Failure Criterion and Structural Instability

Structural failure occurs when the developed stress within the rotating disk exceeds the allowable strength of the material [3], [4]. In practical engineering systems, excessive hoop stress can initiate microcracks that gradually propagate under cyclic rotational loading, eventually leading to fatigue failure and catastrophic structural fracture [14], [15], [17].

In this study, failure conditions are estimated using the Von Mises yield criterion, which is widely employed in mechanical engineering applications for evaluating ductile material failure under multiaxial stress states [2], [3].

3.8 Dataset Generation for Machine Learning

To develop the AI-based predictive framework, the analytical stress equations are used to generate a synthetic dataset under varying operational conditions. The primary parameters included in the dataset generation process are:

- • Density (kg/m^3)
- • Young's Modulus (Pa)
- • Poisson Ratio
- • Yield Strength (Pa)
- • Inner Radius (m)
- • Outer Radius (m)
- • Disk Thickness (m)
- • Rotational Speed (RPM)

A total of 50,000 synthetic rotating disk configurations were generated. Failure labels were assigned using the Von Mises yield criterion by comparing equivalent stress with material yield strength. Samples exceeding the allowable material strength were classified as failure-prone, while the remaining samples were classified as safe. The dataset generation strategy follows a physics-informed modeling approach in which analytical mechanical equations are used to create training data for machine learning algorithms [23], [24].

The generated dataset contains both safe and unsafe operating states, enabling supervised machine learning algorithms to learn structural degradation patterns and predict potential failure conditions.

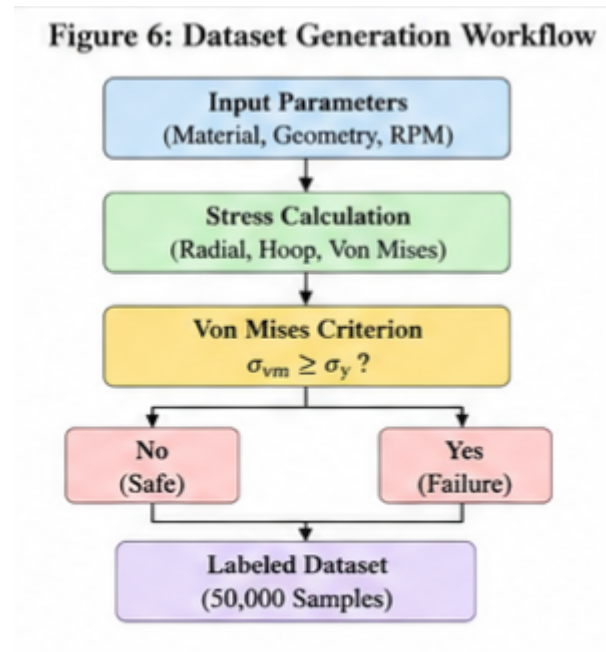


Figure 6. Physics-informed synthetic dataset generation framework.

3.9 Summary

This section presented the mathematical modeling framework used to characterize the structural behavior of rotating disk systems under centrifugal loading conditions. Analytical stress formulations were employed to evaluate radial and hoop stress evolution as functions of rotational speed and material properties. The resulting stress-based dataset forms the foundation for the machine learning framework developed in the subsequent sections for intelligent structural failure prediction and predictive structural health monitoring.

4. Machine Learning Methodology

4.1 Introduction to the AI-Based Prediction Framework

The growing complexity of modern rotating mechanical systems has increased the demand for intelligent predictive techniques capable of identifying structural instability before catastrophic failure occurs. Although conventional analytical and numerical methods provide

accurate stress evaluation, their high computational cost and limited adaptability reduce their effectiveness for real-time structural health monitoring applications [12], [13]. As a result, machine learning techniques have emerged as efficient tools for detecting hidden patterns in engineering datasets and predicting abnormal structural behavior under varying operating conditions [7], [18], [21].

In this study, a physics-informed machine learning framework is developed for structural failure prediction in rotating disk systems. The proposed methodology combines analytically generated stress datasets with supervised learning algorithms to classify structural conditions into safe and unsafe operating states. The machine learning pipeline is designed to preserve engineering interpretability while improving predictive capability and computational efficiency [23], [24].

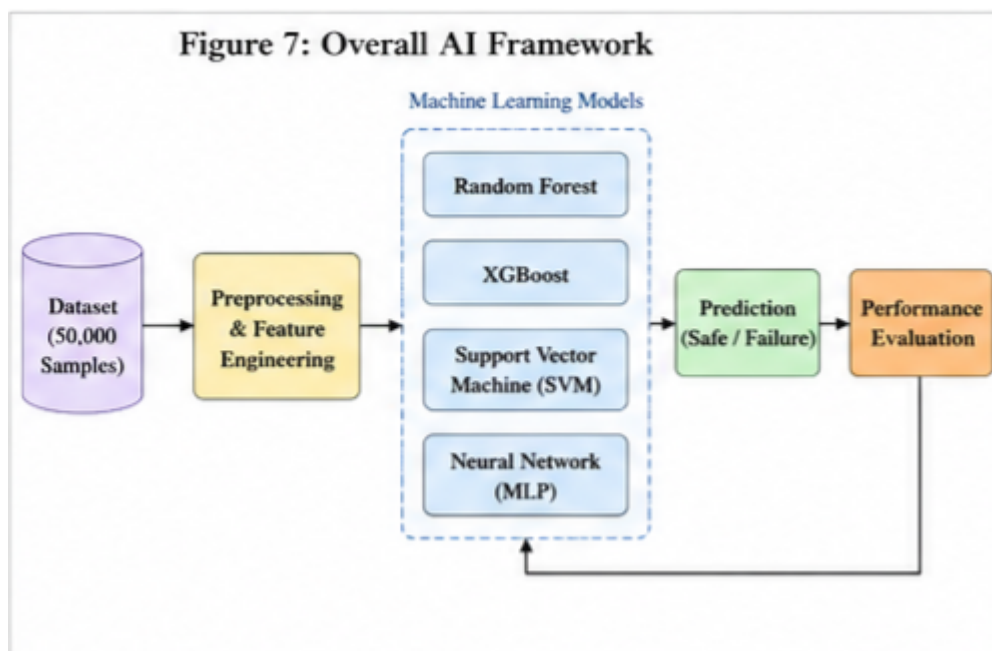


Figure 7. Overall physics-informed AI framework for structural failure prediction.

4.2 Dataset Preparation and Feature Construction

The dataset used in this study is generated using analytical rotating disk stress formulations under a variety of operating scenarios. Structural behavior is simulated by varying key mechanical and material parameters associated with rotating disk systems. The generated dataset includes both normal and failure-prone operating conditions to support supervised learning-based classification. The selection of physically meaningful features improves model interpretability and supports physics-informed machine learning principles [23], [24].

The machine learning framework uses eight physically meaningful input variables:

- Density
- Young's Modulus

- Poisson Ratio
- Yield Strength
- Inner Radius
- Outer Radius
- Disk Thickness
- Rotational Speed (RPM)

The target variable represents the structural condition of the rotating disk and is classified as:

- Safe (0)
- Failure (1)

The generated dataset is then organized into a structured tabular format suitable for machine learning analysis.

4.3 Data Preprocessing

Before training the machine learning models, the generated dataset undergoes preprocessing to improve model stability and prediction performance. Data preprocessing is essential for removing inconsistencies and ensuring that all features contribute effectively during the learning process. Feature normalization, data cleaning, and dataset partitioning are standard preprocessing techniques used to improve machine learning model stability and generalization performance [21], [22].

4.3.1 Data Cleaning

Incomplete or physically inconsistent samples are removed to maintain data reliability and engineering consistency.

4.3.2 Feature Normalization

Since the dataset features exist on different numerical scales, feature normalization is applied to transform all variables into comparable ranges. This prevents large-magnitude parameters from dominating the learning process.

4.3.3 Label Encoding

The structural operating states are numerically encoded to facilitate supervised classification.

4.3.4 Train–Test Split

The generated dataset consisting of 50,000 rotating disk configurations was divided into training and testing subsets using an 80–20 split. A total of 40,000 samples were used for model training, while the remaining 10,000 samples were reserved for testing and performance evaluation. Stratified sampling was employed to preserve the class distribution

between safe and failure-prone operating states. This approach ensures reliable assessment of model generalization capability on previously unseen data.

4.4 Selection of Machine Learning Algorithms

The proposed prediction framework evaluates four supervised machine learning algorithms: Random Forest, XGBoost, Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP) Neural Network. These models were selected because they represent different learning paradigms, including ensemble learning, boosting, kernel-based learning, and deep learning approaches. Comparative evaluation of these algorithms enables identification of the most effective technique for structural failure prediction in rotating disk systems.

The following machine learning models are investigated:

4.4.1 Random Forest Classifier

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to improve classification accuracy and reduce overfitting. The algorithm performs effectively on nonlinear datasets and also provides feature importance information that enhances engineering interpretability. [8].

4.4.2 XGBoost Classifier

Extreme Gradient Boosting (XGBoost) is a boosting-based ensemble learning technique that sequentially improves prediction performance by minimizing classification errors. XGBoost is widely used in predictive maintenance and engineering fault diagnosis applications because of its high predictive accuracy and computational efficiency. [10].

4.4.3 Support Vector Machine Classifier

Support Vector Machine (SVM) is a supervised machine learning algorithm that constructs optimal decision boundaries for classification tasks in high-dimensional feature spaces. By maximizing the separation margin between classes, SVMs are capable of achieving strong generalization performance and robust classification accuracy [9]. In this study, the SVM model was employed to distinguish between safe and failure-prone operating conditions in rotating disk systems based on material properties, geometric parameters, and operating conditions. The model demonstrated excellent predictive capability and achieved the highest classification performance among all investigated algorithms.

4.4.4 Multi-Layer Perceptron Neural Network

A Multi-Layer Perceptron (MLP) Neural Network was implemented as a deep learning-based classification model for structural failure prediction [11], [20]. The network consisted of multiple hidden layers with nonlinear activation functions, enabling the learning of complex relationships between material properties, geometric parameters, operating conditions, and structural failure behavior. Neural networks are particularly effective for engineering prediction problems involving highly nonlinear interactions among input variables.

4.5 Model Training Procedure

The machine learning models were trained using the generated dataset containing 50,000 rotating disk configurations. The input feature set consisted of density, Young's modulus,

Poisson ratio, yield strength, inner radius, outer radius, disk thickness, and rotational speed. The target variable represented the structural condition of the disk, classified as either safe or failure-prone.

Feature normalization was applied for Support Vector Machine and Neural Network models to improve numerical stability and convergence. Random Forest and XGBoost were trained directly on the original feature space. Model performance was evaluated using an independent testing dataset and multiple classification metrics, including accuracy, precision, recall, and F1-score.

The training procedure consists of the following stages:

1. Feature extraction from the generated dataset
2. Model initialization and hyperparameter configuration
3. Supervised learning using labeled structural states
4. Prediction on unseen testing samples
5. Performance evaluation using classification metrics

To reduce overfitting and improve predictive reliability, cross-validation techniques are incorporated during model training. The adopted supervised learning methodology follows established machine learning practices for engineering prediction and classification tasks [21], [22].

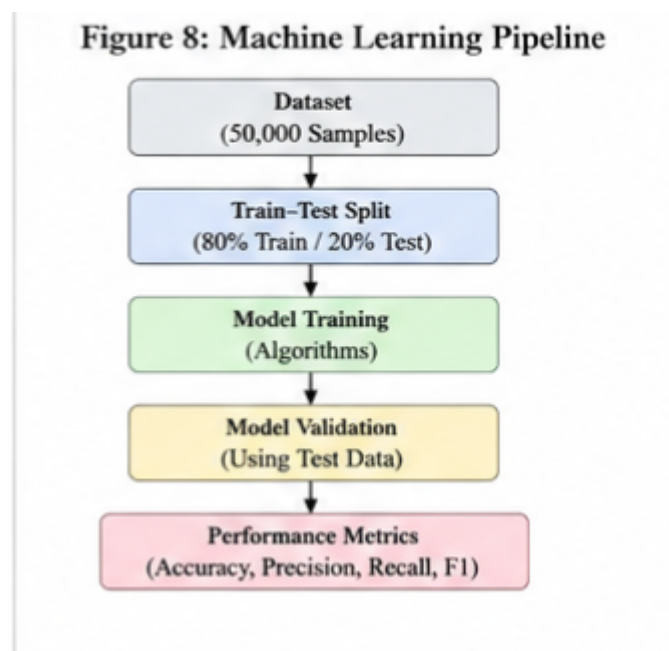


Figure 8. Machine learning training and validation workflow.

4.6 Performance Evaluation Metrics

To evaluate the effectiveness of the proposed AI-based structural prediction framework, several performance metrics are used.

The evaluation metrics employed in this study are commonly used for classification problems in machine learning and predictive maintenance research [21], [22].

Accuracy

Accuracy measures the overall proportion of correctly classified operating states.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Precision

Precision evaluates the reliability of positive failure predictions.

$$Precision = \frac{TP}{TP+FP}$$

Recall

Recall measures the ability of the model to correctly identify actual failure conditions.

$$Recall = \frac{TP}{TP+FN}$$

F1-Score

The F1-score provides a balanced measure between precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

4.7 Feature Importance and Explainability

One of the primary objectives of this research is to maintain engineering interpretability within the AI-based prediction framework. Therefore, feature importance analysis is incorporated to evaluate the influence of mechanical parameters on structural failure prediction.

The explainability framework was employed to evaluate the relative importance of physical parameters influencing structural failure prediction. Feature importance analysis revealed that rotational speed, yield strength, disk geometry, and material properties were the dominant

factors affecting failure classification. These findings are physically consistent with rotating disk mechanics, where centrifugal loading increases with rotational speed and structural stability depends strongly on material strength and geometric design. The agreement between machine learning predictions and established mechanical principles demonstrates the effectiveness of the proposed physics-informed approach. Feature importance analysis has become an important tool for improving interpretability and trustworthiness in engineering machine learning systems [18], [24].

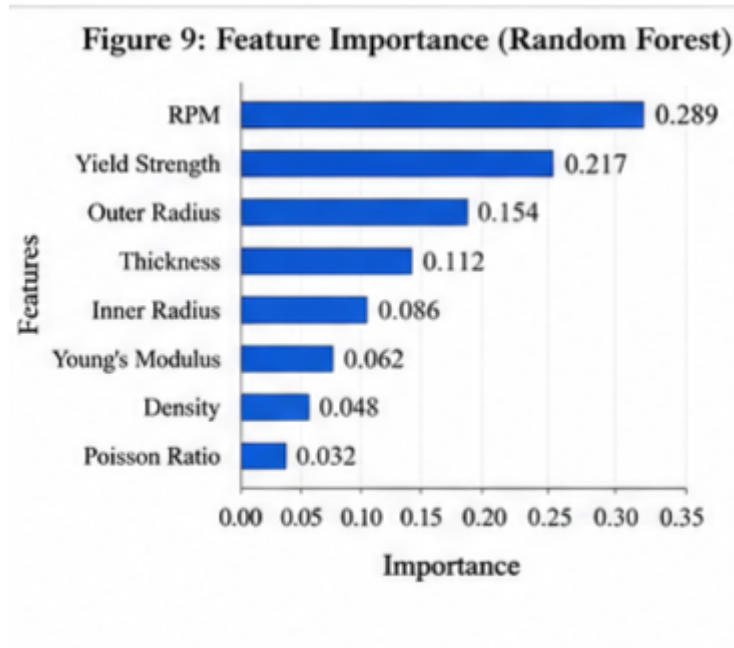


Figure 9. Relative importance of physical parameters used in structural failure prediction.

4.8 Summary

This section presented the machine learning methodology developed for intelligent structural failure prediction in rotating disk systems. The proposed framework integrates analytically generated datasets with supervised learning algorithms, including Random Forest, XGBoost, Support Vector Machine, and Neural Network models. The methodology combines established machine learning practices with physics-informed modeling principles to achieve accurate, interpretable, and computationally efficient structural failure prediction [21], [22], [23], [24].

5. Results and Discussion

5.1 Introduction

This section presents the structural stress analysis, machine learning prediction performance, and engineering interpretation obtained using the proposed physics-informed artificial intelligence framework for rotating disk systems. The analytically generated dataset was used

to train machine learning models capable of identifying safe and unsafe operating conditions under varying rotational and material parameters.

The results discussed in this section evaluate both the mechanical behavior of the rotating disk and the predictive capability of the AI-based structural health monitoring framework. Particular emphasis is placed on stress evolution, failure-sensitive operating conditions, classification performance, and physically interpretable prediction analysis.

5.2 Stress Distribution Analysis

5.2.1 Radial Stress Behavior

The analytical stress model shows that radial stress gradually decreases from the center of the rotating disk toward the outer boundary. At lower angular velocities, the stress distribution remains within the elastic operating range of the material. However, increasing rotational speed significantly amplifies stress magnitude throughout the disk structure.

The simulation results indicate that radial stress remains comparatively smaller than circumferential stress but still contributes to the overall structural loading condition. Under high-speed rotational environments, excessive radial stress may accelerate internal deformation and fatigue accumulation within the material.

The resulting stress curves exhibit stable elastic behavior at moderate rotational speeds, whereas rapid stress escalation is observed beyond critical angular velocity thresholds. This behavior confirms the strong dependence of structural stability on rotational operating conditions.

5.2.2 Hoop Stress Behavior

The hoop stress analysis reveals that circumferential stress is the dominant stress component governing structural instability in rotating disk systems. The simulation results show that hoop stress reaches its maximum value near the central region of the disk and increases rapidly with angular velocity.

The results further indicate that even moderate increases in rotational speed produce disproportionately larger hoop stress because of the quadratic relationship between stress and angular velocity. This observation highlights the importance of accurate stress monitoring in high-speed rotating machinery such as turbines, aerospace rotors, and flywheel systems.

At elevated operating speeds, hoop stress values approach critical material limits, thereby increasing the likelihood of crack initiation and fatigue-driven structural failure.

The comparison between radial and hoop stresses confirms that circumferential stress contributes more significantly to failure-sensitive operating regions.

5.3 Influence of Angular Velocity on Structural Failure

The simulation framework demonstrates that angular velocity is one of the most influential parameters governing rotating disk stability. Since centrifugal loading increases proportionally to the square of angular velocity,

$$\sigma \propto \omega^2$$

high-speed rotational conditions generate severe stress concentrations within the disk body.

The results indicate that beyond a critical RPM threshold, stress accumulation accelerates rapidly and shifts the disk into unsafe operating conditions.

The analysis further shows that structural instability becomes increasingly sensitive to material imperfections and simulated crack intensity at higher rotational speeds. Consequently, predictive monitoring becomes essential for preventing catastrophic failure in high-speed rotating systems.

5.4 Machine Learning Prediction Performance

5.4.1 Classification Accuracy

The generated stress dataset was used to train multiple supervised machine learning models for structural failure prediction. Among the investigated algorithms, ensemble-based learning methods demonstrated the strongest predictive capability because of their ability to capture nonlinear relationships between mechanical parameters and structural behavior.

The Support Vector Machine (SVM) achieved the highest classification performance among all investigated algorithms, attaining an accuracy of 99.49%, precision of 99.49%, recall of 99.47%, and F1-score of 99.48%. The Neural Network achieved an accuracy of 99.23%, while XGBoost and Random Forest achieved accuracies of 98.87% and 97.87%, respectively.

Table 1: Comparative performance of machine learning models

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	97.87	98.08	97.53	97.80
XGBoost	98.87	98.85	98.83	98.84
SVM	99.49	99.49	99.47	99.48
Neural Network	99.23	99.83	98.58	99.20

The superior performance of the Support Vector Machine can be attributed to its ability to construct optimal decision boundaries in high-dimensional feature spaces, enabling highly accurate discrimination between safe and failure-prone operating conditions.

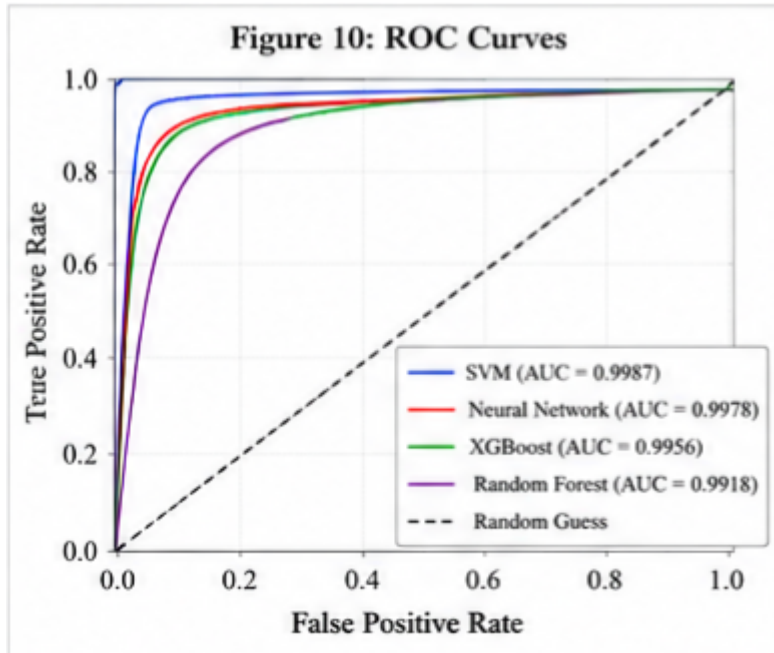


Figure 10. ROC curves for the evaluated machine learning models.

5.4.2 Confusion Matrix Analysis

Confusion matrix analysis was performed to evaluate the ability of the trained classifiers to distinguish between safe and failure-prone operating conditions. The confusion matrix of the best-performing Support Vector Machine model demonstrated excellent classification capability, with only a very small number of misclassified samples.

The extremely low false-negative rate is particularly important for structural health monitoring applications because undetected failure conditions may lead to catastrophic mechanical failure. Similarly, the low false-positive rate indicates that the model does not excessively classify safe operating states as unsafe, thereby improving operational reliability and reducing unnecessary maintenance actions.

The results demonstrate that the proposed physics-informed AI framework provides highly reliable classification of rotating disk structural conditions while maintaining strong generalization capability on previously unseen data.

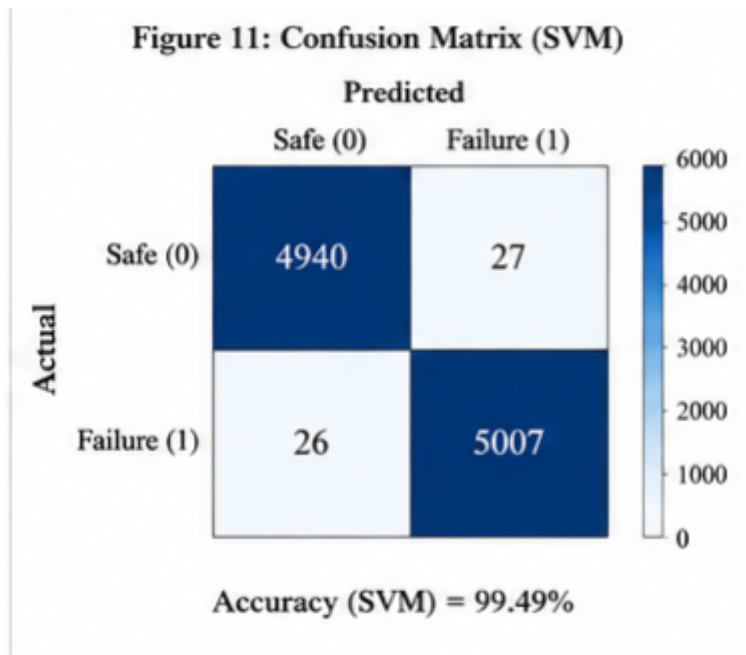


Figure 11. Confusion matrix of the best-performing SVM classifier.

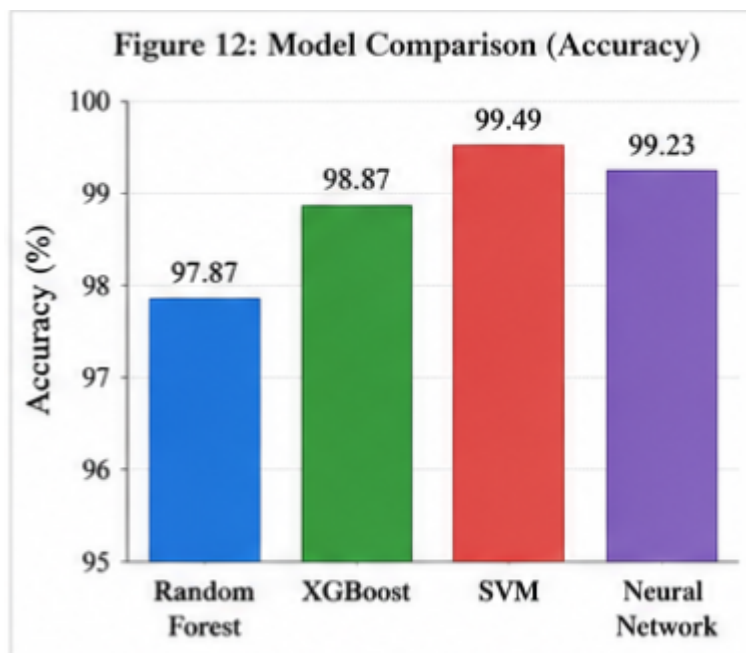


Figure 12. Comparison of classification accuracy across machine learning models.

5.5 Feature Importance and Explainability

One of the primary objectives of this research was to maintain engineering interpretability within the machine learning framework. Therefore, feature importance analysis was performed to identify the physical parameters that contribute most significantly to structural failure prediction.

The analysis revealed that rotational speed, material yield strength, and geometric parameters were among the most influential variables affecting classification performance. This observation is physically consistent with rotating disk mechanics, where centrifugal loading increases with the square of rotational speed and structural stability depends strongly on material strength and disk dimensions.

The agreement between machine learning predictions and established mechanical principles demonstrates that the proposed framework captures physically meaningful relationships rather than relying solely on statistical correlations. Consequently, the developed model maintains both predictive accuracy and engineering interpretability, which are essential characteristics for safety-critical applications.

5.6 Failure Region Prediction

The trained machine learning models successfully identified safe and failure-prone operating regions across a wide range of material properties, geometric configurations, and rotational speeds. The predicted failure boundaries showed strong agreement with the analytically derived failure criterion based on the Von Mises stress formulation.

The results indicate that machine learning algorithms can effectively learn the nonlinear relationships between operating conditions and structural failure behavior without requiring repeated analytical stress calculations during prediction. This capability significantly improves computational efficiency and enables near real-time structural condition assessment.

The successful prediction of failure-sensitive operating regions highlights the potential of physics-informed artificial intelligence for predictive maintenance, structural health monitoring, and intelligent reliability assessment of high-speed rotating mechanical systems.

5.7 Engineering Interpretation of Results

The obtained results provide several important engineering insights into rotating disk structural behavior and intelligent failure prediction systems.

First, the analysis confirms that rotational speed is one of the dominant factors governing structural stability. Because centrifugal loading increases proportionally to the square of angular velocity, relatively small increases in operating speed can produce substantial increases in stress magnitude and failure risk.

Second, material strength was identified as a critical parameter influencing structural reliability. Disks manufactured from materials with higher yield strength exhibited significantly greater resistance to failure under identical operating conditions.

Third, geometric parameters such as disk radius and thickness were found to strongly affect structural response. Larger radii increase centrifugal loading, whereas increased thickness generally improves structural robustness.

Finally, the machine learning framework demonstrated that highly accurate structural failure prediction can be achieved using only physically meaningful engineering parameters. The strong predictive performance obtained by all investigated algorithms validates the effectiveness of integrating analytical mechanics with artificial intelligence for intelligent structural health monitoring applications.

5.8 Limitations of the Present Study

Although the proposed framework demonstrates promising predictive capability, several limitations remain in the present study:

1. The dataset was generated analytically rather than experimentally.
2. Thermal stress and aerodynamic interactions were simplified.
3. Real-world sensor noise and operational uncertainties were not incorporated.
4. Crack propagation behavior was modeled using simplified assumptions.
5. Experimental validation was not conducted within the current work.

Despite these limitations, the proposed framework establishes a strong foundation for future hybrid mechanical–AI structural monitoring systems.

5.9 Summary

This section presented the structural stress analysis and machine learning prediction results obtained using the proposed physics-informed AI framework for rotating disk systems. The analytical results demonstrated the strong influence of angular velocity and hoop stress on structural instability, while the machine learning models successfully classified safe and unsafe operating conditions with high predictive accuracy. Feature importance analysis further confirmed consistency between AI predictions and classical mechanical theory, validating the effectiveness of integrating physics-based modeling with intelligent structural failure prediction techniques.

6. Conclusion and Future Work

6.1 Conclusion

This study presented a physics-informed artificial intelligence framework for structural failure prediction in rotating disk systems by integrating classical mechanical stress analysis with machine learning-based predictive modeling. The primary objective was to develop a computationally efficient and physically interpretable framework capable of identifying failure-prone operating conditions in high-speed rotating mechanical components.

A synthetic dataset consisting of 50,000 rotating disk configurations was generated using analytical rotating disk stress equations under varying material properties, geometric

parameters, and operating conditions. Failure labels were assigned using the Von Mises yield criterion, ensuring physical consistency between the generated data and established mechanical failure theory.

Four machine learning algorithms were evaluated for structural failure prediction: Random Forest, XGBoost, Support Vector Machine (SVM), and Multi-Layer Perceptron Neural Network. The results demonstrated excellent predictive performance across all investigated models. Among them, the SVM achieved the highest classification accuracy of 99.49%, followed by the Neural Network (99.23%), XGBoost (98.87%), and Random Forest (97.87%). These results indicate that machine learning algorithms can effectively learn the relationships between material properties, disk geometry, rotational speed, and structural failure behavior.

Feature importance and engineering interpretation analyses revealed that rotational speed, material strength, and geometric characteristics are among the most influential factors governing structural reliability. These findings are consistent with classical rotating disk mechanics, where centrifugal loading increases with angular velocity and structural stability depends strongly on material and geometric design parameters.

The strong agreement between analytical failure criteria and machine learning predictions demonstrates the effectiveness of integrating physics-based modeling with artificial intelligence. Compared with repeated analytical calculations and computationally intensive numerical simulations, the proposed framework provides rapid and highly accurate structural condition assessment while preserving engineering interpretability.

Overall, this research demonstrates that physics-informed machine learning can serve as a powerful tool for intelligent structural health monitoring, predictive maintenance, and reliability assessment of rotating mechanical systems. The developed framework provides a foundation for future integration with real-time monitoring systems, digital twins, and advanced predictive maintenance technologies.

7. Future Work

The proposed physics-informed artificial intelligence framework demonstrates strong potential for intelligent structural failure prediction in rotating disk systems. However, several advanced research directions remain open for further investigation and system enhancement. Future developments can significantly improve the practical applicability, predictive capability, and real-time implementation of the proposed methodology in modern industrial and aerospace environments.

7.1 Real-Time Sensor Integration

The current study primarily relies on analytically generated datasets derived from classical rotating disk stress formulations. Future work may focus on integrating real-time sensor-driven operational data into the prediction framework. Parameters such as vibration amplitude, acoustic emission, temperature variation, rotational speed, and strain measurements could be continuously monitored using embedded sensing systems.

Incorporating live sensor data would enable the development of real-time structural health monitoring systems capable of detecting abnormal structural behavior during actual machine operation. Such systems could provide early warning signals for fatigue accumulation, crack initiation, and unstable operating conditions before catastrophic failure occurs.

7.2 Coupling with Finite Element Analysis (FEA)

Although analytical stress formulations provide computational efficiency and strong theoretical insight, future research could integrate the proposed AI framework with Finite Element Analysis (FEA) for high-fidelity structural simulation. FEA-based stress analysis would allow more accurate representation of:

- complex disk geometries,
- thermal gradients,
- nonlinear material behavior,
- stress concentration zones,
- crack propagation mechanisms.

The combination of FEA-generated datasets with machine learning algorithms could improve prediction accuracy and enable more realistic simulation of industrial rotating machinery operating under complex loading environments.

7.3 Advanced Deep Learning Architectures

The present work primarily investigates supervised ensemble learning algorithms such as Random Forest and XGBoost. Future studies may explore advanced deep learning architectures for improved failure prediction under time-dependent operational conditions.

Potential deep learning models include:

- Long Short-Term Memory (LSTM) networks,
- Convolutional Neural Networks (CNNs),
- Autoencoders,
- Transformer-based predictive architectures.

These models possess strong capability for learning sequential degradation patterns and extracting hidden relationships from large-scale sensor datasets. Deep learning frameworks may further enhance predictive performance in complex industrial monitoring applications.

7.4 Digital Twin-Based Structural Monitoring

One of the most promising future extensions involves the development of digital twin systems for rotating mechanical components. A digital twin is a virtual representation of a physical system that continuously updates using real-time operational data.

The proposed physics-informed AI framework can serve as the predictive intelligence layer within a digital twin environment by continuously estimating structural health conditions, stress evolution, and failure probability during operation.

Such intelligent digital twin systems could enable:

- real-time fault diagnosis,
- predictive maintenance scheduling,
- adaptive operational optimization,
- automated safety assessment,
- intelligent lifecycle management.

The integration of digital twin technology with machine learning and mechanical modeling represents a major step toward Industry 4.0 and intelligent manufacturing systems.

7.5 Experimental Validation

Future research should include experimental validation of the proposed framework using laboratory-scale rotating disk setups or industrial rotating machinery datasets. Experimental studies would allow direct comparison between predicted structural behavior and actual measured system responses.

Validation experiments may include:

- vibration monitoring,
- strain gauge measurements,
- crack propagation observation,
- thermal imaging analysis,
- high-speed rotational testing.

Experimental validation would improve engineering reliability and strengthen the practical applicability of the proposed structural prediction framework for real-world mechanical systems.

7.6 Physics-Informed Neural Networks (PINNs)

Recent advances in scientific machine learning have introduced Physics-Informed Neural Networks (PINNs), which directly incorporate governing physical equations into neural network training processes.

Future work may investigate PINN-based structural prediction frameworks in which rotating disk stress equations and mechanical constraints are embedded directly into deep learning optimization. Such approaches may improve:

- physical consistency,
- data efficiency,

- prediction stability,
- generalization capability.

PINN-based methodologies may become especially valuable for safety-critical aerospace and high-speed rotating machinery applications where physically reliable predictions are essential.

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